Data Preparation

Modelling + Test

Evaluation

PepcDB

Original DB (28,731)

Crystallizable (5,383)

Non-crystallizable (23,348)

Sequence Similarity < 25%

Sequence Length < 800

Crystallizable (4,736)

Non-crystallizable (20,384)

SwissProt (SP) (604)

Tremble (TR) (2,521)

Sequence Similarity < 25%

Sequence Length < 800

D1

3,846 Crystallizable 19,487 Non-Crystallizable

Training Set

D2

890 Crystallizable 897 Non-crystallizable

Balanced Test Set

148 Crystallizable 89 Non-crystallizable

SP_final

374 Crystallizable 638 Non-crystallizable

TR_final

D1

3,846 Crystallizable 19,487 Non-Crystallizable

TRILL

PLM Classifiers

Crystallizable Protein Generator

Test Prediction

Candidate Crystallizable Proteins

Performance Comparison

Candidate Crystallizable Proteins

PLM Classifiers

Filtered Crystallizable Proteins

Sequence to 2° Structure Compatibility

32 Filtered Crystallizable Proteins

Aggregation Scanner

28 Filtered Crystallizable Proteins

Homologs Removal Coverage ≤ 40% Identity ≤ 35% E-value ≥ 0.5

5 Potential Crystallizable Proteins

3° Structure Prediction & Validation

Reliably Folded Protein Structures

5 Novel Designed Proteins with Computationally Validated 3° Structures

Database

Data

Sequence similarity between SP / TR and D1

Process

Model